

GET 324: Artificial Intelligence and Machine Learning

LABORATORY EXERCISE 10 (MINI-PROJECT) REPORT

Group Information

Group Name: Group C10

Task: Leopard vs Tiger

Team Members

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Project Links

GitHub Repository : <https://github.com/RightFix/big-cat-classifier>

Deployed Streamlit Application: <https://big-cat-classifier.streamlit.app>

Project Report

Dataset Source

The dataset was sourced from Kaggle ('Cats in the Wild Image Classification'). It originally contained 7 distinct classes of wild cats and 3 species of Leopards. We meticulously preprocessed the dataset, discarding the 6 unneeded classes to isolate only the required categories (Leopard and Tiger) for our classification task.

How to Use the Application

Users navigate to the deployed Streamlit web application and upload an image (JPG, PNG) of either a leopard or a tiger. The backend model immediately processes the image and outputs the predicted class along with a confidence score.

Challenges Encountered

During development, our primary challenge was model optimization. We initially designed and trained a custom Convolutional Neural Network (CNN). However, the model failed to learn the complex distinguishing features between leopards and tigers, resulting in poor accuracy. We also spent considerable time iteratively cleaning the dataset to ensure no data leakage from the discarded classes.

Solutions and Improvements

To resolve the custom model's poor performance, we pivoted to a Transfer Learning approach. Utilizing a pre-trained feature extraction architecture allowed the model to leverage existing learned weights, which drastically improved both our validation accuracy and convergence speed. Possible future improvements include applying heavy data augmentation to handle varied lighting and backgrounds, and implementing real-time webcam capture on the Streamlit interface.